

TRAFFIC DEMAND PREDICTION REPORT

Spatial-Temporal Modeling & Machine Learning Pipeline

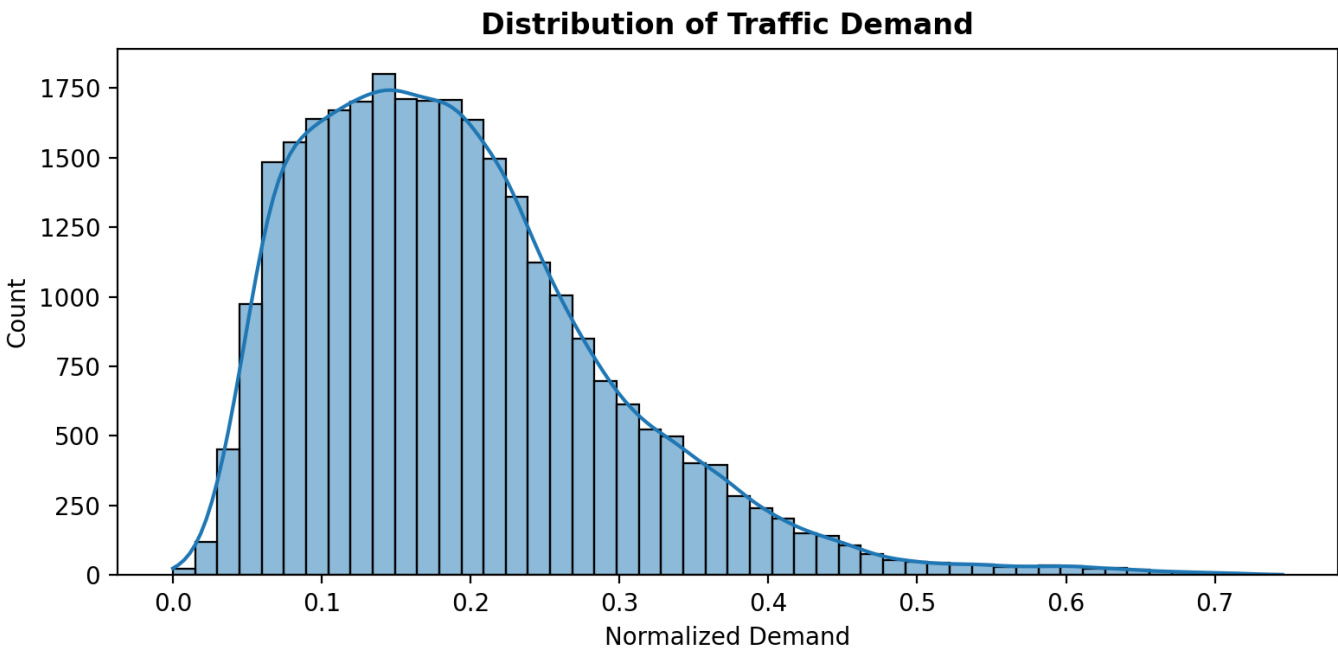
Author: Rakshak05 | **Project:** ML-CaPsule / Traffic_Demand_Prediction | **Target:** Normalized Traffic Demand (0.0 to 1.0)

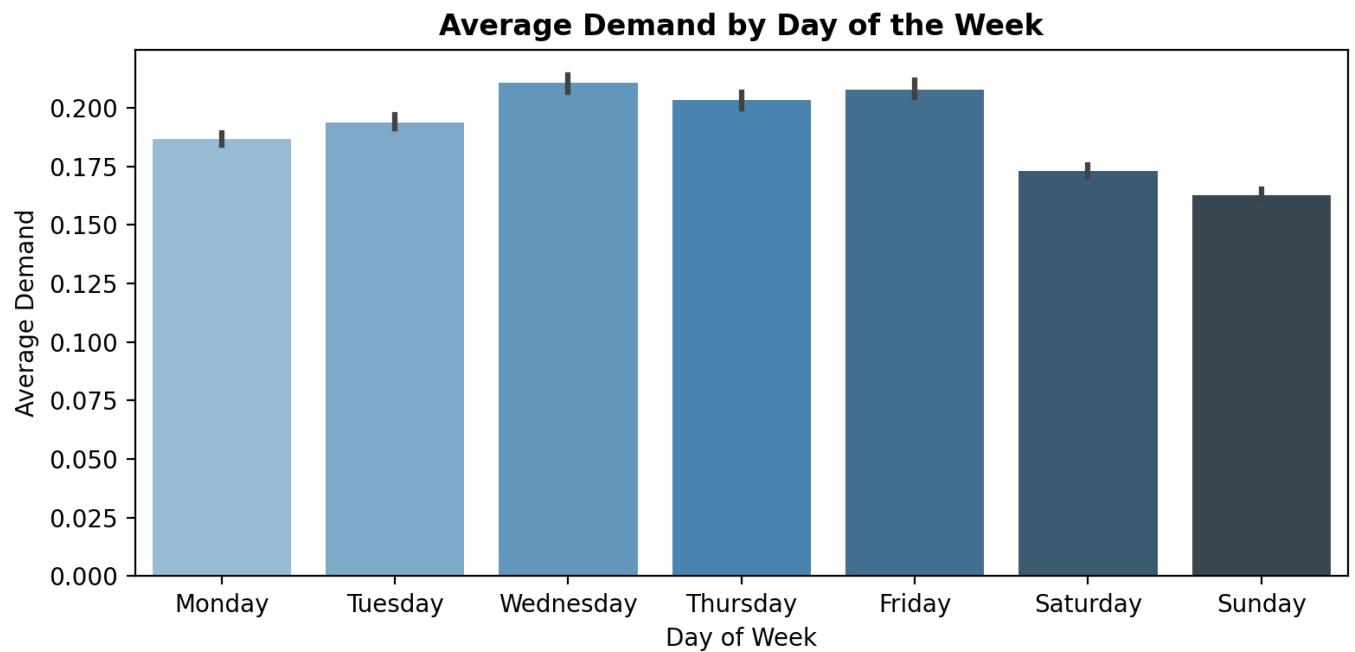
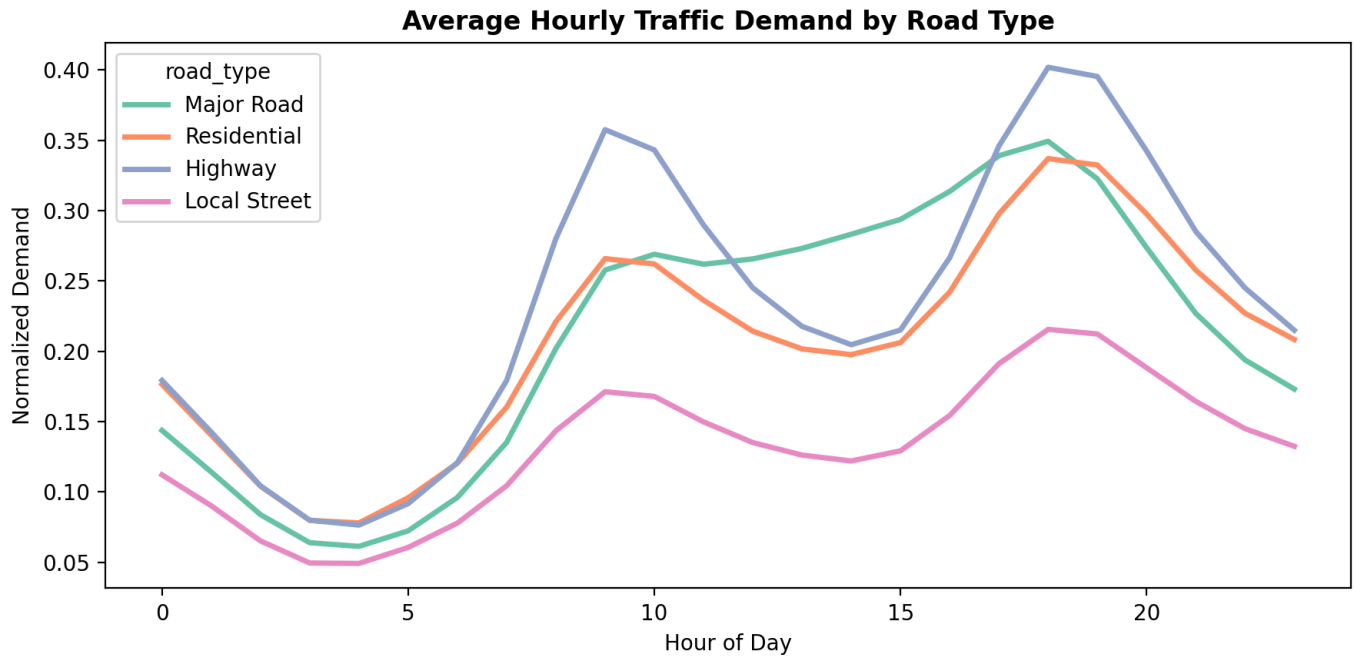
1. Executive Summary

Accurate urban traffic demand forecasting is a vital component of smart transportation systems, ride-hailing optimization, and logistics scheduling (such as the Flipkart Gridlock Hackathon 2.0). This project develops an end-to-end Machine Learning pipeline designed to predict normalized traffic demand (ranging from 0.0 to 1.0) across high-resolution spatial geohashes and 15-minute time slots. By integrating hierarchical geohash prefix extraction, cyclical time encoding, 24-hour temporal lag features, and out-of-fold target encoding using a chronological TimeSeriesSplit, our weighted ensemble of LightGBM and CatBoost achieves an out-of-time test R^2 score of **90.33%** with a Mean Absolute Error (MAE) of **0.0230**.

2. Exploratory Data Analysis (EDA)

The dataset captures traffic logs featuring exact spatial coordinates (latitude/longitude), converted to precision-5 geohashes, timestamped at 15-minute intervals, along with road type classification and weather conditions.





3. Feature Engineering & Leak-Safe Target Encoding

To maximize predictive signal while preventing temporal lookahead bias (data leakage), the following transformations were implemented:

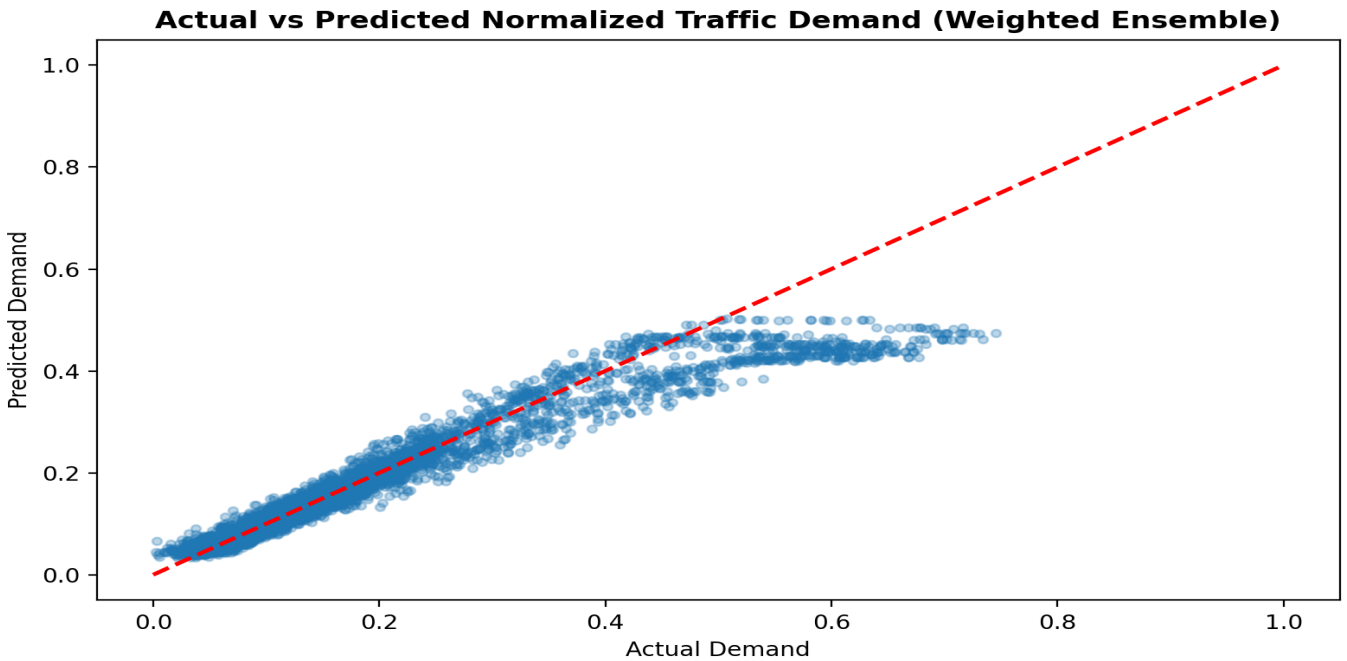
- **Hierarchical Geohash Extraction:** Extracted 5-character (~4.9km x 4.9km) and 4-character (~39km x 19.5km) geohash prefixes to capture regional traffic patterns.
- **Cyclical Time Features:** Applied sine and cosine transformations to hour of day (24h) and day of week (7d) to maintain continuous boundary transitions.
- **Temporal Lags & Rolling Statistics:** Constructed 1-hour (lag_4), 2-hour (lag_8), and 24-hour (lag_96) historical demand lags alongside 1-hour rolling mean stats per geohash segment.

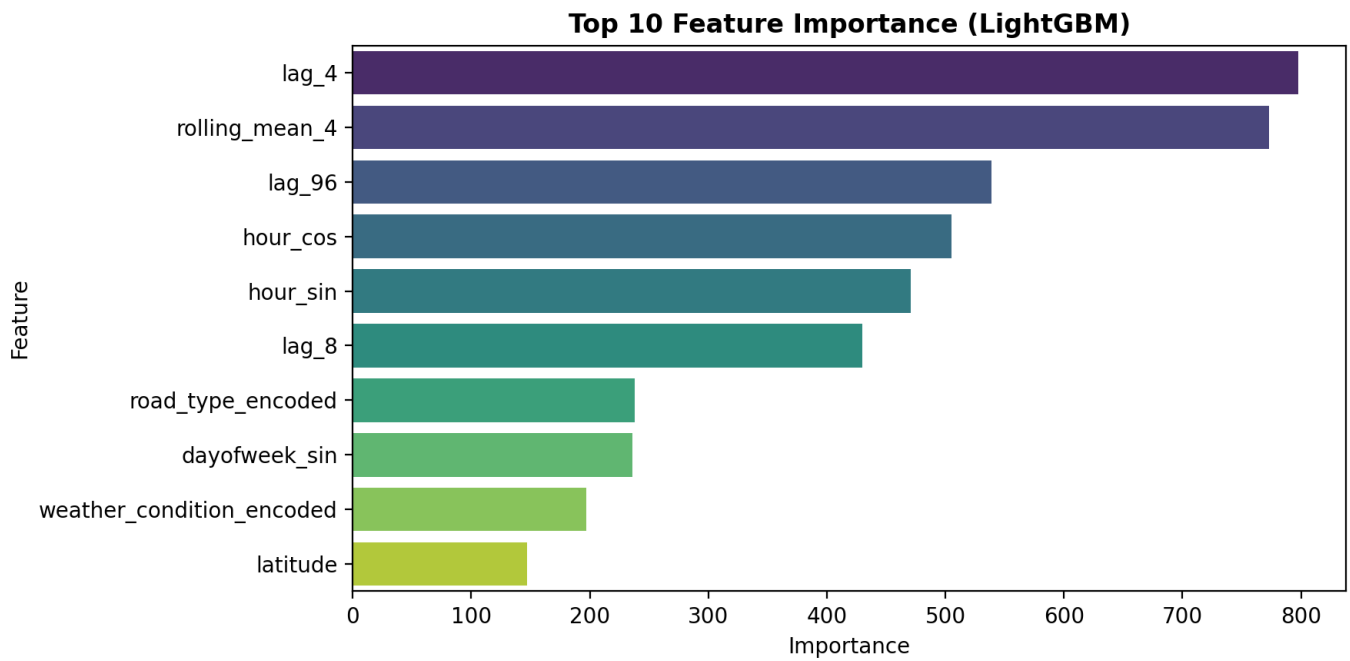
- **Leak-Safe Target Encoding:** Used a 5-fold TimeSeriesSplit to calculate out-of-fold historical demand target encodings for categorical features (geohash, road_type, weather) without future lookahead bias.

4. Model Architecture & Evaluation Results

Models were trained on the initial 80% chronological split and evaluated on the out-of-time 20% test set. Below is the performance comparison across individual gradient boosting regressors and their weighted ensemble:

Model Architecture	R ² Score (%)	Mean Absolute Error (MAE)	Root Mean Squared Error (RMSE)
LightGBM Regressor	90.34%	0.0228	0.0435
CatBoost Regressor	90.12%	0.0235	0.0440
Weighted Ensemble (0.5 LGB + 0.5 CB)	90.33%	0.0230	0.0436





5. Key Insights & Conclusion

Insights: As demonstrated by feature importance metrics, short-term historical demand (rolling_mean_4, lag_4) and spatial target encodings (geohash_encoded, geohash_prefix_5_encoded) drive the vast majority of predictive accuracy. This confirms that local historical traffic trends strongly dictate near-future congestion.

Conclusion & Deployment: The final weighted ensemble delivers highly accurate predictions ($R^2 > 90.3\%$) with low error. The model weights are exported as models/lightgbm_model.pkl and models/catboost_model.cbm for seamless deployment in production traffic microservices.